# Analysis 1: Mock Mid-semester Examination

**Time:** 3 hours **Marks:** 100

Answer *any four* questions. Each question carries 25 marks.

You may use that $\mathbb{R}$ satisfies the l.u.b. axiom. You may also use the definitions of the limit of a sequence, limsup and liminf of a sequence, subsequence of a sequence, convergence of a series, and continuity and uniform continuity of a function.

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I. Let $(a_n : n \geq 1)$ be a bounded sequence. For $n \geq 1$ let $E_n = \{a_k : k \geq n\}$ and let $t_n = \text{l. u. b. } E_n$.

(a) Show that there is $t \in \mathbb{R}$ such that $\lim_{n\to\infty} t_n = t$.

(b) Show that there is a subsequence $(b_k : k \geq 1)$ of $(a_n : n \geq 1)$ converging to $t$.

(c) Suppose that $(a_n : n \geq 1)$ is Cauchy. Show that it converges to $t$.

II. Let $(a_n : n \geq 1) \subseteq (0, \infty)$ be a sequence and for $n \geq 1$ let $b_n = \max\{a_j : j \leq n\}$ and $c_n = a_n/b_n$.

(a) Suppose $\lim_n c_n = \theta < 1$. Show that $\lim_n a_n = \alpha < \infty$.

(b) Suppose $\limsup_n c_n = \theta < 1$. Show that $\limsup_n a_n = \alpha < \infty$.

(c) Suppose $a_n \neq a_m$ whenever $n \neq m$, and $\lim_n c_n = 1$. Give examples to show that $\limsup_n a_n$ could be finite or infinite.

III. Suppose the series $\sum_k a_k$ and $\sum_k b_k$ are convergent. Let $c_k = a_k b_k$ for $k \geq 1$.

(a) Give an example to show that $\sum_k c_k$ may not converge.

(b) If in addition $a_k \geq 0$ and $b_k \geq 0$ for all $k$, show that $\sum_k c_k$ converges.

IV. Let $A \subseteq \mathbb{R}$ and let $f, g$ be bounded continuous functions from $A$ to $\mathbb{R}$.

(a) Suppose $g(x) \neq 0$ for all $x \in A$. For $x \in A$ let $h(x) = f(x)/g(x)$. Show that $h$ is continuous.

(b) Suppose $A = [0, 1]$. Show that $f$ is uniformly continuous.

(c) Suppose $A = (0, 1)$. Give an example to show that $f$ need not be uniformly continuous.

V. Let $(a_n : n \geq 1)$ be such that $\limsup_n \sqrt[n]{|a_n|} = 2$. For $x \in \mathbb{R}$ let $f_n(x) = \sum_{k=1}^{n} a_k x^k$.

(a) Show that $f_n(x)$ converges for $|x| < \frac{1}{2}$.

(b) Show that $f_n(x)$ does not converge for $|x| > \frac{1}{2}$.

(c) For $|x| < \frac{1}{2}$ let $f(x) = \lim_n f_n(x)$. Show that $f$ is continuous on $\left(-\frac{1}{2}, \frac{1}{2}\right)$.